TEAM 06

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Lasso-Logistic Regression and Artificial Neural Network for Prostate Adenocarcinoma Metastasis Prediction and Biomarker Discovery Based on Gene Expression Profile

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Abstract

Prostate adenocarcinoma is a leading cause of cancer-related deaths, often due to late detection and metastasis. Limited prognostic tools and high false-negative rates contribute to its high mortality. This study utilizes RNA-seq data from TCGA, consisting of 52 normal adjacent tissues, 592 non-metastatic, and 99 metastatic samples, to identify metastasis-related biomarkers using machine learning models. We performed a two-step feature selection process: first, through differential gene expression (DEG) analysis, and second, using a machine learning-based approach. As a result, **MYH11, PRMT1, and TENT4A** were identified as potential biomarkers using **Lasso-Logistic Regression**. Based on these biomarkers, we developed an **Artificial Neural Network** classifier that achieved the best performance. These three biomarkers are associated with metastasis induction, playing roles in **cytoskeleton disorganization, epithelial-mesenchymal transition regulation, and mRNA stabilization**. This study is expected to serve as a milestone for future applications in Indonesia.

Materials and methods

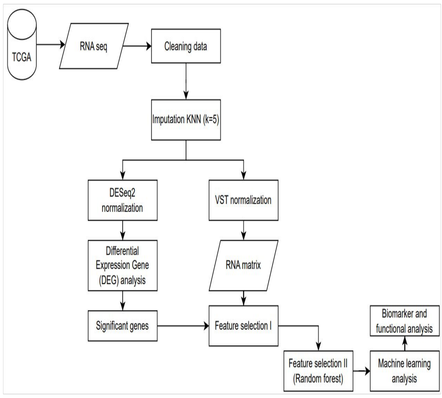

Figure 1. RNA-seq analysis pipeline, from preprocessing and normalization to feature selection, classification, and biomarker identification.

Results and Discussion

1. Feature selection I : Differential Gene Expression (DEG)

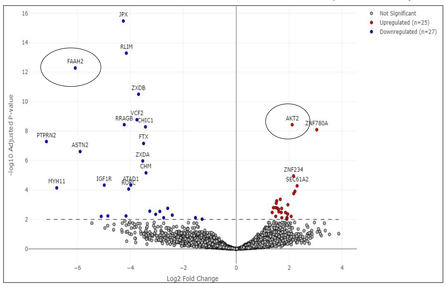

Figure 2. Volcano plot of differentially expressed genes for metastatic and non-metastatic prostate cancer patients

- Several genes are significantly (q<0.01) **down-regulated** (27 genes) and **up-regulated** (25 genes) in metastatic prostate cancer
- **AKT gene (up-regulated):** Promotes bone metastasis
- **FAAH gene (down-regulated):** Acts as a tumor suppressor, primarily in lungs

2.Data distribution (PCA)

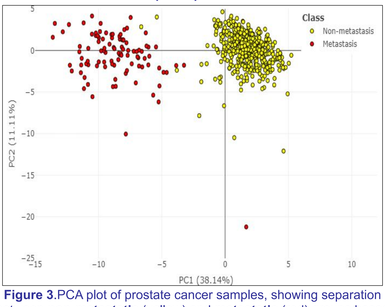

Figure 3.PCA plot of prostate cancer samples, showing separation between **non-metastatic** (yellow) and **metastatic** (red) groups based on gene expression profiles.

- The PCA plot shows distinct gene expression patterns between **metastatic** (red) and **non-metastatic** (yellow) prostate cancer samples.
- The clear separation of data points indicates **significant differences** in their **gene expression profiles**.

3.Feature selection II : Feature importance

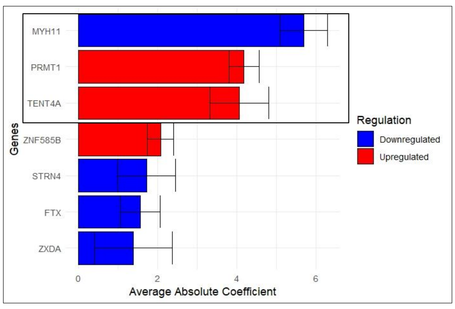

Figure 3. Feature importance graph for 7 key genes in metastatic prostate cancer

- Trained **Lasso-logistic regression** and **random forest** to assess feature importance, with **logistic regression** showing the best performance (100%)
- Identified seven key genes, selecting three as biomarkers for the next classifier model (**MYH11, PRMT1, TENT4A**).

4.Cross validation & external validation

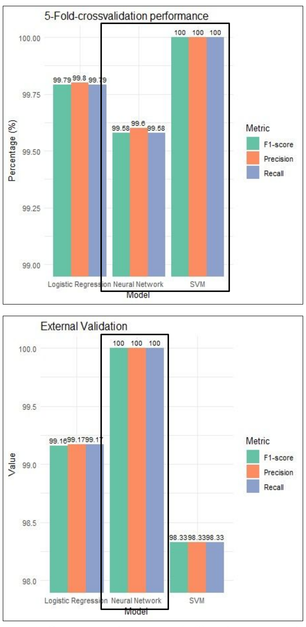

Figure 4. Cross validation model (left); external validation model (right)

- SVM performs well in cross-validation but suffers from **overfitting**.
- **External validation** is more important, and the **neural network** is the best at generalizing patterns, making it the **preferred choice**.

5.Biomarker pathway analysis

- **MYH11 (Myosin Heavy Chain 11)** = downregulated in metastasis, leading to cytoskeleton disorganization and easier cancer cell detachment from the primary tumor
- **PRMT1 (protein arginine methyltransferase 1)** = upregulated in metastasis, EMT induction by regulating E-cadherin and ZEB1, promotes motility and invasion by β-catenin downregulation
- **TENT4A** (Terminal Nucleotidyltransferase 4A) = upregulated in metastasis, regulates mRNA stability and degradation, potentially influencing cancer progression by modulating oncogene expression and RNA quality control.

Conclusion

- **Lasso-Logistic Regression** outperformed Random Forest in feature selection.
- **Artificial Neural Network** achieved high classification performance during external validation using only three biomarkers.
- The three identified biomarkers **(MYH11, PRMT1, and TENT4A)** are associated with metastasis induction through **cytoskeleton disorganization, EMT promotion, and mRNA stabilization.**

Limitation

- This study relies on RNA-seq data from non-Indonesian patients due to limited available patient data in Indonesia.
- Requires more external validation, which can be achieved by increasing the dataset size.

REFERENCES